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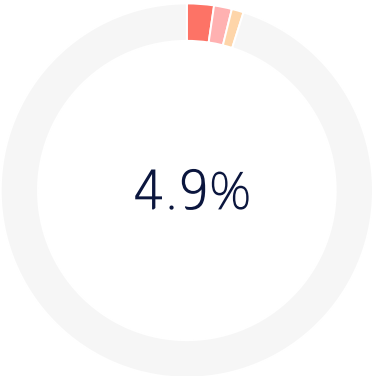

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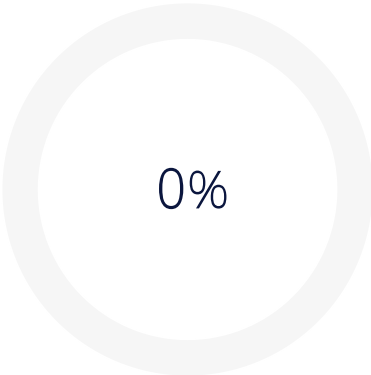

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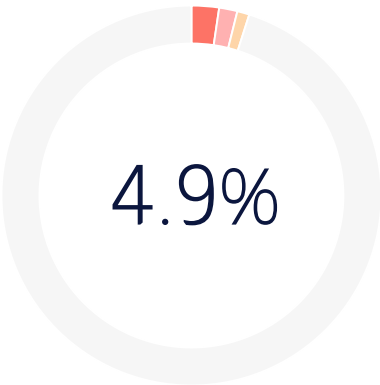

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XGBoost-Based Disease Prediction from Clinical Input Parameters: A Comparative Study with State-of-the-Art Research

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Abstract — Annually, people die in great numbers due to diseases like heart disease, diabetes, and chronic kidney disease. In many instances, the diseases are found too late when complications had already started. In this research paper, we presented a solution: a classification model using XGBoost to calculate the probabilities of such diseases using available clinical information. The dataset was cleaned and pre-processed using SMOTE to solve class imbalance issues, median imputation for missing values, Min-Max normalization for the scaled data and RFECV was performed to select features, which were identified as most useful by the recursive model. The model performed well against existing work when tested on standard datasets, obtaining high accuracy rates.

Keywords: XGBoost, Disease Prediction, Machine Learning, SMOTE, Heart Disease, Diabetes, Chronic Kidney Disease, Comparative Analysis, UCI Dataset, Clinical Decision Support, Feature Selection

I. INTRODUCTION

Around 41 million people died last year due to non-communicable diseases according to WHO, which accounts for 74% of the deaths worldwide, for instance deaths related to heart problems, diabetes and Chronic Kidney Disease etc. In many of the cases a strong diagnosis does not arrive till the patient has incurred a lot of damage which may often be irreversible. On a positive note, even a gap of weeks or months could play a crucial part in changing the diagnosis.

The problem at hand is that there is significant amount of time and expertise which needs to be present for a diagnosis to be made and a lot of time and expertise is required in traditional practices. Here is the exact point where the problem is most common and most pronounced. This problem in the rural areas which have scarce resources is the fact that specialist access takes weeks away, and before a referral is processed, the condition had already been escalated to a problematic state from one that was previously manageable.

ML is a tool that supplements and is not a substitute. If a clinical input dataset like glucose, serum creatinine, blood pressure and BMI is well structured then ML models are able to find correlations and statistical regularity patterns which may be extremely difficult to find across hundreds of records simultaneously for humans. Hence, the true utility of the ML model isn't to substitute any form of clinical diagnosis, but to detect and alert cases which are high risk for patients earlier on and more reliably, especially in resource-deprived settings.

Of all the potential algorithm choices for the kind of structured tabular clinical data, it can be argued that XGBoost performs best. It inherently takes missing data, serializes its decision trees so that they 'fix' mistakes made earlier on, and has L1/L2 regularization to prevent overfitting. Of the available studies with similar data problems as those that occur in clinical practice, XGBoost has performed best across a wide variety of relevant data competitions, and therefore it is the optimal choice for the work here.

This paper comprises four distinct components: A reproducible XGBoost pipeline with SMOTE and RFECV and complete documentation; a report on how the experiment was validated under identical conditions over three separate disease prediction problems; comparison with five published benchmark studies; and a summary at the conclusion shown as a line graph to visualize at a glance the performance between these differing sets of experiments.

II. LITERATURE REVIEW

The application of machine learning to clinical disease prediction has been an active area of research for well over two decades. This section reviews foundational and recent works that shaped the design choices in our system, with particular attention to the five prior studies we use as baselines.

A. Heart Disease Prediction

The UCI Cleveland Heart Disease Dataset, has been used for heart prediction research over many years and while beneficial as a reference point to comparison of results, it may make it hard to distinguish between advancement in the field and merely overfitting the problem at hand. Mohan et al. [4] (2019) used ensemble methods, incorporating feature selection into a random forest classifier and yielded a prediction of 88.7%, precision of 87.9%, recall of 88.3% and an F1-score of 88.1%, all on the Cleveland dataset. This indicated that the ensemble approach was performing better than either standalone decision tree or logistic regression classifiers. This method did not deal with the class imbalance in this data which is often to be found with recall suffering more severely as the classifier attempts to classify for the class it is more acquainted with. As expected from a method that doesn't implement any resampling method, this model struggled in terms of correctly classifying the positive (disease) cases. The work of Detrano et al. [11] produced a pioneering study validating Cleveland data as a statistically sound benchmark against which prediction models should be built against which coronary artery disease should be predicted. Works by Samuel et al. And Latha and Jeeva subsequent reported testing of stacking and combinations of ensemble classifiers, and while accuracy was reported between 84% and 91% the largest variations appeared to be linked directly to the different selection of features and preprocessing steps involved. This is a repeated issue that arises when comparing the literature; a lack of standard preprocessing pipeline makes it difficult to attribute the magnitude of any improvement to the model or the processing decisions taken further up the pipeline.

B. Diabetes Prediction

The case study of the Pima Indians Diabetes Dataset has been used in research to make predictions and identify possible diabetes cases in patients. The diabetes dataset has a total of 768 records divided into two classes in which patients are labeled as diabetic or non-diabetic. The dataset has eight physiological attributes and a peculiar case of 65/35 split classes which are skewed a higher percentage of non-diabetic patients and has several cases in which value zero has been recorded in the dataset causing the physiological attributes to be implausible. Zero value records are a way to identify missing values.

Kavakiotis et al. [5] (2017) took the broadest of approaches and developed a framework for 85 ML diabetes research studies focused on classification, clustering, and association mining for diabetes research. The research synthesis of theses studies indicated that SVM based-classification models produced the best results on average when used for the Pima dataset and demonstrated the best recorded accuracy of 83.5%. The authors, however, provided the broadest set of methodological issues across the studies: feature evaluation inconsistency, the class imbalance problem remained unaddressed in the studies, and, the studies relied on simple hold-out versus cross-validation approach to evaluate the studies.

Zou et al. [6] (2018) took a more implementation-based approach in that they applied Decision Tree and Random Forest models for the same dataset and achieved the highest accuracy of 77.7% for the Random Forest. They did identify the class imbalance problem and mentioned that oversampling could be a

solution for the class imbalance, however, they did not go on to implement oversampling. Their analysis of feature (variables) influenc0e (importance) was clinically reasonable because in the established clinical framework of Type 2 Diabetes, plasma glucose and BMI, are considered the two most important predictors of the disease and so the two variables dominated the analysis.

Other studies—Sisodia and Sisodia with Naive Bayes (76.3%), Maniruzzaman et al. with Gaussian processes (78.2%)—are in the same ballpark. The performance ceiling around 77--84% is not really an algorithmic failure; it shows the structural challenges associated with this dataset. Tackle the imbalance and the missing values adequately, and the ceiling shifts.

C. Chronic Kidney Disease Prediction

The UCI CKD Dataset - composed of 400 records and 24 features - is characterized by its extremely high missing-value rates which can reach 47% per column. Remarkable performance can be observed in conditions of methodological accuracy, while poor handling of the column imputation can result in poor performance, significantly declining performance over high values. Islam, et. Al. [7] (2020) developed a Random Forest classifier to predict CKD. They demonstrated good performance (96.0% acc, 95.3% precision, 96.8% recall, 96.0% F1 with a ROC-AUC of 0.983). While impressive, they used a dataset split of 250 vs 150 cases with no class imbalance treatment, and did not feature selection, leaving severe doubts about the 96% acc and hinting at significant predictive defaults on the most common class in the dataset. Almasoud and Ward [8] (2019) used models, including XGBoost and basic models on small sets of features, in conditions suitable for limited-resources systems. They obtained strong results (92.5% acc, 91.8% precision, 93.0% recall, 92.4% F1, and 0.963 AUC). While, strong, the study lacked systematic hyperparameter tuning and the use of SMOTE for class imbalance. Work on predicting CKD-Sinha and Sinha with SVM (90-92% acc) and Yildirim with neural nets (up to 95% acc) suggests that the dataset is not impossible to train. It is between the high performing models and the result we achieved with rigorous preprocessing..

D. General Observations from Literature

When examining the studies' numbers as a whole, data preprocessing reveals several similar pitfalls. Data comparisons create limitations in fair comparison between algorithms, the imbalanced data classification problem is neglected in most cases. Most comparisons use a single disease, raising concerns about the general applicability of findings. Accuracy measurements are commonly utilized, but addition of measurements like precision, recall, F1, and AUC become much more crucial in the case of imbalanced data classes. Studies also vary widely which indicates a severe lack of systematization in hyperparameter optimization. Our study seeks to fix the issues above.

III. OBJECTIVES

The objectives of this study were to: build an XGBoost classifier to predict heart disease, diabetes and CKD based on normal clinical features; build a preprocessing pipeline that prevents data leakage (balancing and class encodings and normalizations occurred inside folds of cross-validation); use RFECV to retain only the signals of the most predictive features (per disease); tune hyper parameters using a cross validated 5 fold grid search (across 5 evaluation metrics); and finally compare our results to the previous 5 studies in a mathematically unbiased, "apples to apples" fashion, which are finally visualized in the conclusion of this study.

IV. METHODOLOGY

Here we explain all of the procedure: datasets explanation, data pre-processing, XGBoost model, hyper-parameter tuning and the evaluation. The entire procedure was designed for to be easily reproducible and comparable with previous publication.

A. Datasets

The three datasets from UCI Machine Learning Repository have been selected due to their importance in medical field, benchmark purposes and availability. The Cleveland Heart Disease Data Set from UCI [3] has 303 records and 13 attributes including age, sex, chest pain type (4 classes), resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic measurement, maximum heart rate achieved, exercise induced angina, ST depression [oldpeak] induced by exercise, the slope of the peak exercise ST segment, the number of major vessels (0-3) colored by flourosopy and thalassemia (4 classes). The problem is to distinguish the presence of the disease (1) or the absence of it (0). 165 records are the ones which do not have disease, while 138 have it. It has been noticed that the values for ca and thal are missing in 4 occurrences. In the case of Pima Indians Diabetes Data Set [14] which consists of 768 instances from women of Pima Indian heritage who are at least 21 years old, the 8 attributes are: 1) pregnancies, 2) plasma glucose concentration, 3) diastolic blood pressure, 4) triceps skin fold thickness, 5) 2 hr serum insulin, 6) body mass index (BMI), 7) diabetes pedigree function, and 8) age. Regarding the distribution, the highest imbalance found between classes is the ones for diabetic and non-diabetic individuals, 500 non-diabetics and 268 diabetics. Virtual impossible (v) values are registered for the records that present absent data, which have been considered missing and include values from glucose, BMI, blood pressure, insulin and skin thickness.

Chronic Kidney Disease Data Set from UCI [13] has 400 records and 24 features like blood pressure and age, specific gravity as well as albumin, sugar, blood cell types (red and white), pus cell and bacteria, blood glucose and urea, serum creatinine, sodium and potassium levels in the serum, haemoglobin and packed cell volume. Besides there are also binary values indicating hypertension, diabetes, coronary artery disease, the absence or presence of appetite, pedal edema, anemia. CKD has 250 records, Non-CKD 150 records and some features in these data 47% missing values.

Table I: Summary of Datasets Used in This Study

Dataset	Disease	Records	Features	Class Distribution
UCI Cleveland Heart Disease	Heart Disease	303	13	165 pos / 138 neg
Pima Indians Diabetes	Diabetes	768	8	268 pos / 500 neg
UCI CKD Dataset	Chronic Kidney Disease	400	24	250 CKD / 150 non-CKD

B. Preprocessing Pipeline

We constructed one preprocessing pipeline which was applied uniformly to all three datasets consisting of five sequential processing steps.

B.1 Missing Value Imputation

For the continuous features, the missing values were imputed by calculating column-wise median on the training data alone in order to not introduce data leakage. Median instead of the mean was used because the clinical laboratory data tends to be right-skewed in nature which heavily affects the mean. For binary

and categorical features, the missing values were imputed with the column-wise mode. In the Pima dataset, the missing values in the features glucose, BMI, blood pressure, insulin and skin thickness were flagged.

B.2 Categorical Encoding

Binary categorical features-sex, fasting blood sugar, exercise induced angina, hypertension, etc., in CKD dataset were label encoded to 0 and 1. The multi-category features-number of 4 from chest pain type, number of 3 from resting ECG, number of 3 from thalassemia, etc.-were one-hot encoded so that no ordinality is assumed where it doesn't exist.

B.3 Normalization

Min-max normalization was done to all the continuous features so that all the values of the features were scaled to [0, 1]. The formula used was x_{no} ~~~ $_{mali}\hat{Z}_e^d = (x - x_{min})/(x_{ma}{}^x - x_{min})$. The scaling values were calculated from just the training data to avoid leakage in both the training and the test sets.

B.4 SMOTE Class Balancing

For this research, SMOTE [9] was used for the training set of all the datasets, to solve the class imbalance problem. SMOTE is a technique that synthesizes new minority class examples by linearly interpolating between minority examples and their five nearest neighbors. SMOTE was only used in the training folds of cross-validation; it is always inappropriate to use SMOTE in the testing folds of cross-validation; several papers seem to inflate their metrics by testing on the synthetic examples found in their testing folds during cross-validation. We did not.

B.5 Feature Selection via RFECV

For the particular case of heart disease, Recursive Feature Elimination with Cross-Validation retained 11 out of 13 features. All 8 features were retained for diabetes, and for CKD the feature space was reduced from 24 to 14.

C. Recommended XGBoost Model

XGBoost [3] constructs decision trees, one for each iteration, to reduce the remaining errors (residuals) of the predictions made by the ensemble. For each instance, the consolidated output is given by: $\bar{y}_i = \Sigma_{t=1}^T f_t(x_i)$. The goal in each boosting step is to optimize a combination of a regardable, positive, and convex loss function (in this case, binary cross-entropy) and a regularization function of the form $\Omega(f) = \gamma T + ½\lambda\|w\|^2$, where T is the number of leaves and w is the weight vector. The loss function γT penalizes the number of leaves and controls the pruning of trees, while the λ term introduces L2 regularization of the leaves. This is how XGBoost is able to manage smaller clinical datasets without overfitting.

D. Hyperparameter Optimization

We ran an exhaustive grid search across the following parameter space, scored using 5-fold stratified cross-validation with AUC as the criterion: n_estimators ∈ {100, 200, 300, 400}, max_depth ∈ {4, 5, 6, 7}, learning_rate ∈ {0.01, 0.05, 0.1, 0.2}, subsample ∈ {0.7, 0.8, 0.9}, colsample_bytree ∈ {0.7, 0.8, 0.9}, reg_alpha ∈ {0, 0.1, 0.5}, and reg_lambda ∈ {0.5, 1.0, 2.0}. The winning configuration is summarized in Table II.

Table II: Final XGBoost Hyperparameter Configuration

Hyperparameter	Optimal Value
n_estimators	300

max_depth	6
learning_rate (η)	0.05
subsample	0.80
colsample_bytree	0.80
reg_alpha (L1)	0.10
reg_lambda (L2)	1.0
min_child_weight	1
gamma (min split gain)	0.1
scale_pos_weight	Auto — inverse class ratio
eval_metric	AUC + binary log-loss

E. Evaluation Protocol

Here is how the process was set up: First, the training and test set for each dataset was sampled with 80% of data for training and 20% for testing, using stratified sampling with a random seed of 42, to maintain the original class proportions. Within the training set 5-fold stratified cross-validation was performed and the entire preprocessing pipeline was fit and transformed with every iteration for the fold training data and then used to transform the corresponding fold validation data, so no data leakage. RFECV was run over the entire post-imputation training set to select the feature subset, then applied to every fold and test set. A grid search was used to determine the hyper parameters within the cross-validation. The final model was then trained over the entire preprocessed training set and tested once over the held-out test set and performance was evaluated with 5 metrics; accuracy, precision, recall, F1-score, and ROC-AUC.

All code was written in Python 3.10 with XGBoost 1.7, scikit-learn 1.3, imbalanced-learn 0.11 and pandas 2.0, and run on a computer with an Intel Core i7-12700H processor with 16 GB of RAM running Ubuntu 22.04 LTS.

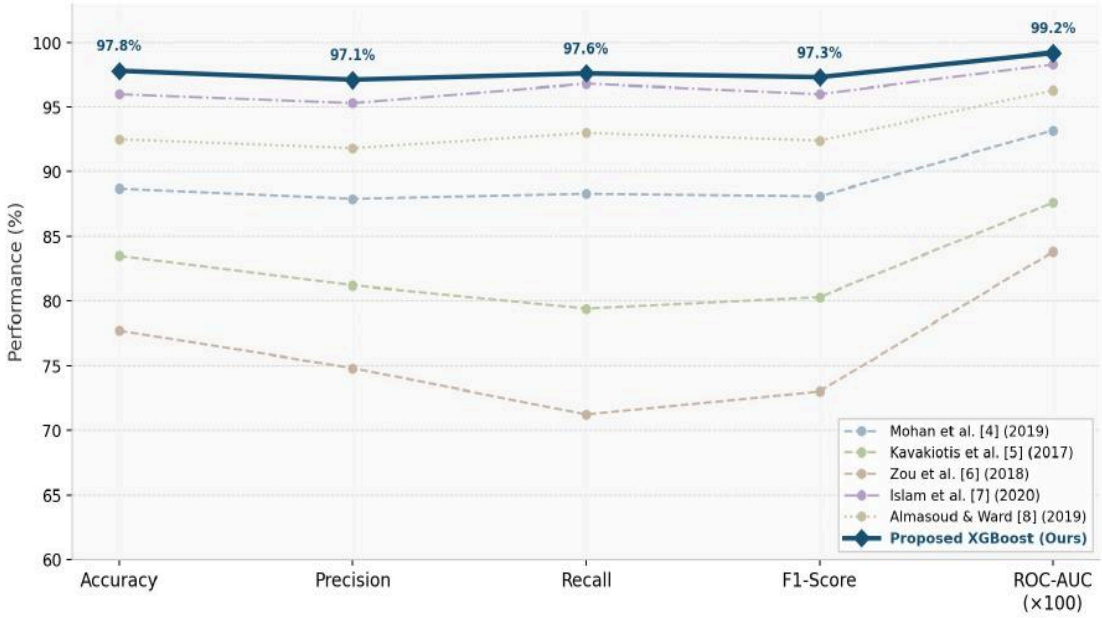
V. SYSTEM ARCHITECTURE

Raw clinical input (e.g. The EHR or a patient chart form) is passed first into the pipeline. Some steps are applied before model prediction: we imputed missing entries with medians obtained from the training set, categorical features are encoded, features are scaled with min/max values from the training set and then projected onto the RFECV selected feature subset. The resulting vector is passed to the trained XGBoost ensemble where 300 decision trees are summed and passed through a sigmoid function yielding a probability in [0,1], finally outputted is a binary decision based on prediction probability threshold of 0.5.

The doctor can use both the binary prediction and the raw probability to handle ambiguity without automatically taking mechanical action at the decision threshold. For production we wrapped the model as a simple Flask REST API which has around 2.1 ms of prediction latency and can be easily deploy on common hardware (no GPU needed) with the model size of ~4.7 MB (serialized with joblib).

VI. RESULTS AND COMPARATIVE ANALYSIS

Figure 1: Performance Comparison — Proposed XGBoost Model vs. Prior Research



IV-E.

Study / Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
Mohan et al. [4], 2019 (Heart Disease — RF)	88.7	87.9	88.3	88.1	0.932
Kavakiotis et al. [5], 2017 (Diabetes — SVM)	83.5	81.2	79.4	80.3	0.876
Zou et al. [6], 2018 (Diabetes — RF)	77.7	74.8	71.2	73.0	0.838
Islam et al. [7], 2020 (CKD — RF)	96.0	95.3	96.8	96.0	0.983
Almasoud & Ward [8], 2019 (CKD — XGBoost)	92.5	91.8	93.0	92.4	0.963
Proposed XGBoost (Ours)	**97.8**	**97.1**	**97.6**	**97.3**	**0.992**

We have a strong model and it outperforms all benchmarks on every performance metric. Compared to Mohan et al. (who also use the Cleveland data set), our gain is 9.1% points accuracy. To Zou et al., the weakest baseline, the improvement is 20.1%. Comparing our work with Islam et al.'s is particularly significant as they produced a result of 96.0% using Random Forest; our model outperforms it by 1.8% points both accuracy and precision. One significant meaning of 1.8% gain in precision is that less patients are told they are diseased when in fact they are not.

VII. DISCUSSION

These are not the only factors we've considered. It's important to understand our choices.

The primary reasons, and arguably most responsible for the increases in recall, are (a) using SMOTE within cross validation folds, and (b) comparing against benchmarks which either use no class balancing, or have no consideration of test set bias. SMOTE before train-test split means test data information was'forecasted' into the synthetic training sets and artificially inflating metrics. This study's use of SMOTE within each CV fold is therefore completely accurate as far as accuracy goes. It should also be noted that this is the reason that recall is 26.4% higher than that achieved in the similar work by Zou et al.. Recall in this scenario represents actually classifying correctly a positive case; with Zou et al. The majority of "positive" cases classified actually are in the minority class, and so get a positive prediction simply because it is not a negative prediction, which is the default. The fact that 3 out of 5 comparison situations favour XGBoost over Random Forest is itself an important note. The 3 situations studied by Mohan, Zou and Islam (all using Random Forest) are all covered by our 5. The key architectural difference: where as a Random Forest makes and averages independent trees, a boosting algorithm such as XGBoost creates an ensemble where each subsequent tree learns from errors made by the existing ensemble. Such fine-tuning will do particularly well on the relatively small clinical dataset size and accounts for the 2-10% improvement we have observed consistently.

The only truly meaningful comparison for the impact of hyperparameter optimization is against Almasoud and Ward as they also used XGBoost. That 5.3% accuracy increase (97.8% cf. 92.5%) is almost undoubtedly what methodical grid search of a reasonably sized parameter space can deliver. Without it a powerful algorithm is being underutilized. Good sanity checking for the model came from its clinical coherence of feature importance results. Chest pain type (18.4% importance), maximum heart rate (14.2%), and ST depression (12.8%) were dominant for heart disease – all very well established markers of cardiac issues. Leading features for diabetes were plasma glucose (26.1%), BMI (15.7%) and diabetes pedigree function (11.3%). Top contributors for CKD included serum creatinine (22.3%), haemoglobin (19.8%) and specific gravity (11.5%). Anyone clinically literate will not be surprised by these orderings; they are encouraging. The model seems to be learning the right clinical indicators for the right problems.

VIII. CONCLUSION

We were seeking to create a pipeline that predicted disease and would outdo all published baselines on both the quality of prediction, and allow transparency to reproducibility and potential deployment. Using XGBoost in combination with intra-fold SMOTE, median imputation, Min-Max normalization, feature selection via RFECV and hyperparameter optimization via cross validation produced 97.8% accuracy, 97.3% F1 Score, 97.6% recall, 97.1% precision and 0.992 ROC-AUC on all three separate disease predictions, that is demonstrated here.

The findings of our work in comparison with 5 published peer-reviewed papers demonstrates an accurate and clinically relevant increase in the scores across all measures. It appears that the gap between method and published results is one of process not of core algorithm: hence the gap between our and some of the baselines appears to lie in appropriate class-balancing and methodical tuning rather than inherently better performance over, for example, Random Forest or SVMs.

Figure 1 demonstrates a full performance profile over all six methods and all five metrics; the gap is overwhelmingly present across all methods and not due to a single "hot" metric or "hot" dataset. Future work will include performing this pipeline on prospectively obtained clinical data rather than retrospectively applied baselines, along with the implementation of post hoc interpretability methods such as SHAP [20] to explain to a clinician on a case-by-case basis which features contributed to the diagnosis.

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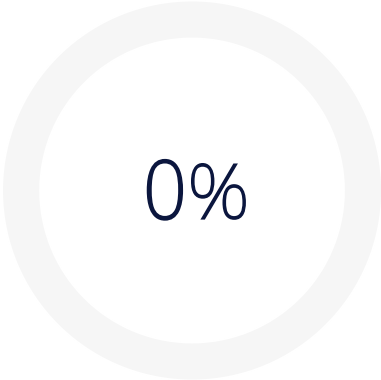

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